

Statistical and Machine Learning SS26

Exercise Sheet 4

Setup

Install and load the `faraway` package, and load the `fat` dataset. We use `brozek` (percent body fat estimated by Brozek's equation) as the response variable

Inspect the dataset: How many observations and predictors does it contain?

How do we find out what exactly the variables measure?

1) Split data into training and test sets

Split the data into a training set (70%) and a test set (30%)

2) Ordinary Least Squares

- a) Fit a standard linear regression model using all predictors on the training set
 - b) Compute the test MSE
 - c) Look at the summary output. Are all predictors statistically significant?
 - d) What potential issues do you notice?
-

3) Lasso regression with fixed λ

Fit a Lasso regression on the training set using $\lambda = 0.03$ (Use package `glmnet`)

4) Ridge regression with fixed λ

Fit a Ridge regression on the training set using $\lambda = 0.03$

5) Comparision Ridge - Lasso

- a) Compare the Ridge coefficients to the Lasso coefficients
 - b) Why does Lasso produce exact zeros while Ridge does not?
 - c) Explain the terms shrinkage and sparsity
-

6) Select λ by cross-validation

- a) Why is choosing the right λ important?
 - b) Use `cv.glmnet()` to select the optimal λ for both Lasso and Ridge via 10-fold cross-validation
 - c) Plot the CV error curves
 - d) What are the optimal λ values (`lambda.min`) for each method?
-

7) Variable selection with Lasso

- a) Using the optimal λ from cross-validation, inspect the Lasso coefficients
 - b) Which variables does the Lasso select? Does this make intuitive sense given the nature of the data?
-

8) Compare all three models

- a) Compute the MSE for OLS, Ridge (λ from 6d), and Lasso (λ from 6d), summarize the results in a table
- b) Which model performs best on the test set? Is the difference large?
- c) Consider the bias-variance trade-off from the lecture - how does it explain what you observe?